AES Student Recording Competition
Nashville 2026

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AES Member Number: #837099

Category 2: Studio Recording

Title: Samba Funk

Instruments: Electric Guitar, Piano, Drums, Acoustic Bass, Percussions
Recording Format: 48kHz/24bit
AES Submission Format: 48kHz/24bit
Dates: recorded Oct 2024; mixed Feb–Apr 2026

Recording equipment:
DAW: AVID Pro Tools
Console: SSL Origin (32-ch)
Preamps: direct to ADC, no processing on input
Monitoring system: Genelec 8030C w/ Genelec sub (near field), Yamaha NS10 w/ JBL (near field), and PMC (room)

Mixing equipment:
DAW: AVID Pro Tools
Monitoring system: PMC (studio room), Kali Audio LP-6 (Atmos room), Sony MDR7506 (headphones)
Plugins: Avid Channel Strip, Ozone 12 Equalizer, D3 ExpGate, D-Verb; NI Raum; Chow Tape Model; Teletronix LA-2A (hardware)

This project took place at Belmont University under the supervision of Dr Wesley A Bulla. I was responsible for recording, editing, and mixing.

Concept:
Samba Funk was composed by Paulo Olivera, a professor in the College of Music & Performing Arts at Belmont University, who performed it with his colleagues as part of the Belmont University School of Music Faculty Jazz Quartet. The session took place at Robert E. Mulloy Studio A in Belmont's Barbara Massey Rogers Center.

The goal for this recording was to capture the energy and groove of the piece: a tight, punchy rhythm section with a forward-driving samba-funk feel. Every source was recorded cleanly and directly, to give the fullest possible control to shape character, balance, and space during the mix.

Recording:

The session made full use of the studio's live room and its surrounding isolation booths, giving each source its own acoustic space while keeping the players in close visual contact.

The layout was arranged around communication and separation. Because the guitar is the primary voice of the piece, and was being played by the composer himself, it was placed nearest to the control room, so that we could relay better communication about how the recording was sounding and if/when we needed to do any punch ins. Because the guitar was captured via DI, it was kept out in the main room alongside the drums, while the guitar amp was placed in a separate room letting us mic it cleanly in isolation. The drums occupied a large portion of the main room and were surrounded by baffles to control spill into the other sources. For the overheads, a Glyn Johns configuration was used where one microphone was positioned above the snare and the second out by the low tom. Both were kept equidistant from the snare so the kit images cohesively and stays phase-aligned around a centered snare. This technique was chosen to provide a vintage, natural drum sound. The acoustic bass was isolated in a booth on the right side of the room, the grand piano in a separate booth on the left. And adjacent to the piano, the percussionist played from another booth, switching between instruments across the piece.

The percussion setup introduced a challenge: because the player was interchanging instruments (congas, shakers, etc.) throughout the take, we had to adjust levels manually while recording to set an appropriate gain for each piece of percussion as it came up.

Source	**Microphone**	**Pattern**
Kick (in)	Neumann FET U47	Cardioid
Snare top	Shure SM81	Cardioid
Snare bottom	Shure SM81	Cardioid
Hat	Neumann KM-84	Cardioid
Hi tom	AKG 414	Hyper-cardioid
Low tom	AKG 414	Hyper-cardioid
Overhead 1 (Glyn Johns #1, over snare)	Neumann U87	Cardioid
Overhead 2 (Glyn Johns #2, by low tom)	Neumann U87	Cardioid
Percussion	Sony C48	Cardioid
Acoustic bass	AKG C12 VR	Cardioid

Source	Microphone	Pattern
Piano low	Neumann U87	Cardioid
Piano mid	AKG 480 w/ CK61ULS	Cardioid
Piano hi	AKG 480 w/ CK61ULS	Cardioid
Guitar amp	Shure SM57	Cardioid
Guitar DI	Countryman DI	—

Figure 1: Recording Microphone List



Figure 2: View of the studio from inside the control room

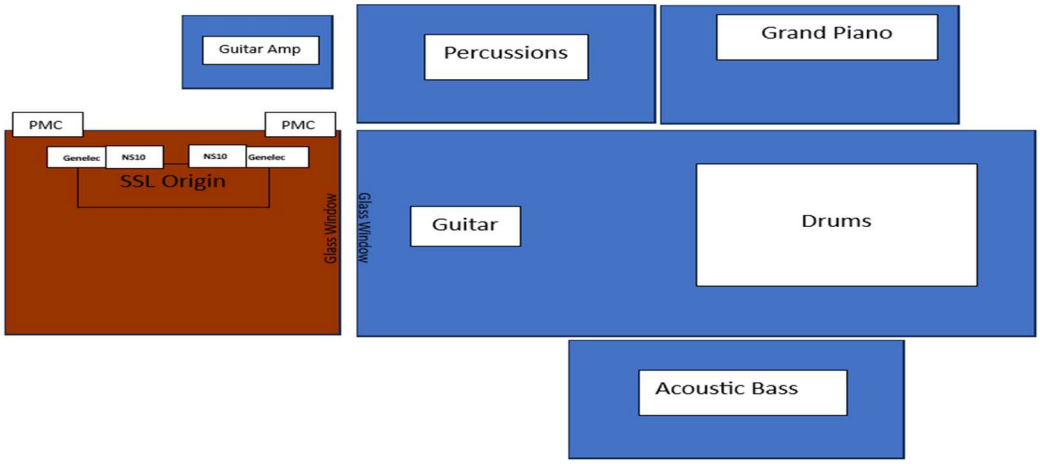


Figure 3: Session Layout

The edit and mix:

The mixing stage was carried out both at home and in the studio, moving between a few different monitoring systems to make sure the mix translated well. PMC monitors in the studio room, Kali Audio LP-6s in the Atmos room, and Sony MDR-7506 headphones at home for detailed checks. Throughout the process I went back and forth with my supervisor on each element of the mix, refining one part at a time until the whole thing sat together and sounded cohesive.

At first, I went overboard editing the drums, trying to remove any and all bleed between the different drum microphones. My supervisor pointed out that this was working against the character of the piece: because the song leans into a vintage, retro jazz vibe, a little bleed between the mics is a good thing rather than a bad thing. So instead of aggressively editing the drum tracks, I took a lighter approach and placed gates on the high and low tom tracks to control them without stripping out the natural spill. One track that did need attention was the hi-hat, which had an impurity in the sound, a squeaking noise, so I removed it entirely; there was more than enough hi-hat present in the overheads to carry it. I also separated the percussion into separate tracks, one per instrument being played, so that any processing I applied would act on that specific instrument rather than colliding with the others sharing a single track.

To give the drums a sense of space, I created a new stereo aux track for a drum reverb and added a small-room verb using D-Verb, set to a 501 ms decay and 87% diffusion.

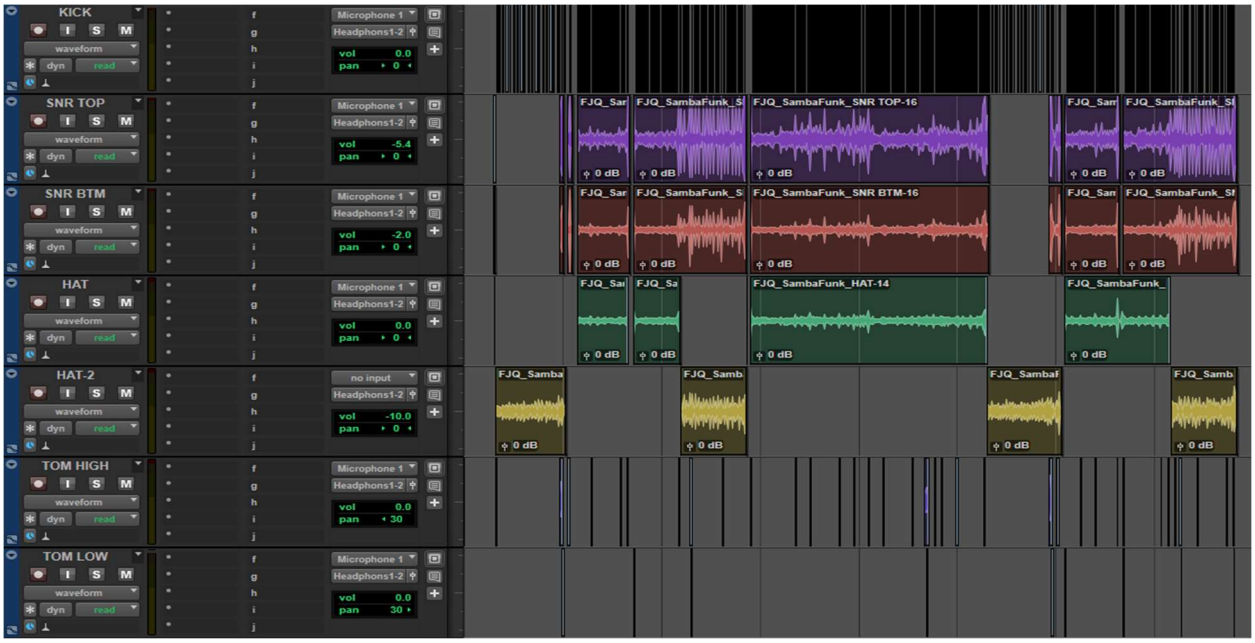


Figure 4: Aggressive editing of the drums, this was discarded later and reset to the original tracks.

The next round focused on EQ across all the elements. I added a notch to the tracks (as shown in Fig. 5) to remove the boxiness that was building up in the low mids, cleaning up so the parts stopped competing for the same space. I also applied gentle compression with the Avid Channel Strip on the kick and bass to firm up the low end. To finish the tonal shaping, I chose to incorporate tape saturation on the master fader. Using a tape saturation can lend both warmth and brightness to a mix, which were the exact characters I was looking for this style of music.



Figure 5: EQ on Guitar Amp and Kick track showing notch filtering around 500Hz



Figure 6: Tape Saturation settings using Chow Tape Model

The final stage of the mix was improving the guitar. My supervisor suggested reaching for an analog component instead of a plugin, utilizing an LA-2A, to bring out more character in the guitar. Running the track through it added the character I wanted, but it introduced a problem: it raised the noise floor of the whole track. To mitigate this, I edited out the silent sections of the guitar so the raised noise floor wouldn't be heard between phrases. Then I re-EQ'd the track and added reverb using the Raum plugin, set to a 2s decay with a 6.27 ms predelay, to make the guitar sound full and sit naturally in the mix.

Evaluation

Making this recording taught me both technical and collaborative nature of mixing. The clean, direct capture at the tracking stage gave me the flexibility I needed later, and the iterative back-and-forth with my supervisor shaped nearly every decision in the mix. The most valuable lesson was learning when not to fix something and pulling back from over-editing the drum bleed showed me that technical cleanliness and musical character aren't always the same goal, and that a little imperfection can serve the vintage aesthetic that might be necessary for the music.

I also came away with a better instinct for problem-solving in the mix. Reaching for analog-style processing added the warmth and character I wanted, but created problems such as the raised noise floor from the LA-2A and was a reminder that every processing choice carries a trade-off. Working through it with EQ and reverb, rather than abandoning the effect, taught me to shape a sound toward the result I was after instead of settling for the first version. Overall, I am pleased with how the final mix balances a clean, controlled foundation with the analog warmth and looseness that give the piece its groove.